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# Data Lab 2026

QC in PLINK2 | GWAS | Meta-Analysis

# Software to be used: PLINK2.0 alpha

- A new version of PLINK: a free, open-source whole genome association analysis tool
  - Association tests (GWAS), genetic data preprocessing, post-GWAS processing
  - For GWAS: supports quantitative (continuous) and dichotomized qualitative (binary) traits
  - Not for genotype calling or result visualization
  - Not taking genetic relatedness into account: need to remove related individuals if any
- Available on UPPMAX HPC
- More reference: https://www.cog-genomics.org/plink/2.0/

# UPPMAX HPC

- High-Performance Computing: using supercomputers and computer clusters to solve advanced computational problems.
- UPPMAX: The computer center of Uppsala University, part of NAISS (The National Academic Infrastructure of Supercomputing in Sweden)
- Three clusters:
  - Bianca: sensitive data, e.g., real patient data, not connected to internet
  - **Rackham:** for local projects with PIs at UU, and for courses
  - Snowy: for courses and some local projects
- https://docs.uppmax.uu.se/

# Log in to Rackham/Snowy

- Most convenient: From the laptop's terminal with SSH:
  - `ssh <username>@rackham.uppmax.uu.se`
  - Enter your password
  - Use regular Linux commands
  - The lab materials: `/crex/proj/uppmax2024-2-1/DATA_LAB/data`
- Website: ThinLinc (https://rackham-gui.uppmax.uu.se/)
  - Need to set up a two-factor authentication
  - Not necessary
- https://docs.uppmax.uu.se/getting_started/login_rackham/
- https://docs.uppmax.uu.se/cluster_guides/snowy/
- Download results from Rackham:
  - `sftp <username>@rackham.uppmax.uu.se`
  - `get <filename>` (assumed it is at your home directory: /home/<username>)
  - To upload: `put <filename>`
  - To logout: `exit`

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# Handling big datafiles and jobs

- When you log in, you are at the “login node”:
  - It is shared
  - Only do short and light things: editing, deleting, moving files
    - Start interactive session
    - Schedule jobs (sbatch)
- Jobs
  - Performing computational jobs with large datafiles and abundant codes
  - Developing computational scripts with many line, for large datafiles
- Interactive session (write command lines one-by-one)
  - `interactive -A uppmax2024-2-1 -M snowy -t 2:00:00 -n 2`
    - Using 2 cores (no more than 16) for 2 hours
  - When you finish, type `exit` to quit the interactive nodes
- Job scheduler (when you have prepared a shell script, and when your job will take hours)
  - `sbatch -M snowy -A uppmax2024-2-1 your_analysis.sh`

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# Job scheduler (not likely to be used in this lab)

- In your .sh script:

```
#! /bin/bash -l
#SBATCH -A <project-number>
#SBATCH -p core
#SBATCH -n 2
#SBATCH -t 3:00:00
#SBATCH -J <your-job-name>
#SBATCH --mail-type=ALL
#SBATCH --mail-user=<your-uppmax-account-email>

module load <packages-you-need>

some_path_variable="/crex/proj/uppmax2024-2-1/DATA_LAB/data"

plink2 <analysis-you-will-be-run> <arg>
Rscript <an-R-script.R>
```

You can then access by ${some_path_variable} in following lines

Example plink commands in later slides

- You can write this offline and paste to your editor on Rackham:
  - `nano your-analysis.sh`
  - When you finish, press keys ‘ctrl’ and ‘o’, then ‘Enter’; and ‘ctrl’ and ‘x’

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# Software for Lab on Rackham

- PLINK2:
  - `module load bioinfo-tools`
  - `module load plink2`
  - Note: If you are running against a binary phenotype, don't use this latest version on UPPMAX: plink2/2.00-alpha-5-20230923
  - Instead: `module load plink2/2.00-alpha-3.7-20221024`
  - There were fatal bugs for binary phenotype in that version
  - Check version: `plink2 --version`
- R:
  - `module load R_packages/4.2.1`
  - Type `R` to enter the coding interface

# Data (Don't edit anything in the course folder)

- Location: `data_loc="/crex/proj/uppmax2024-2-1/DATA_LAB/data"`
  - You will be able to use a shortcut ${data_loc} after typing the line above
  - What's inside the directory: `ls ${data_loc}` Note: if your filename contains space, must add quotes "${data_loc}"
  - PLINK1 files for a quick view of how genotype files look like: .ped and .map files in tab- or space-delimited formats, could be visualized by text editors, or simply:
    - `head ${data_loc}/test.ped | column -t` To make it visually pleasing by aligning the columns in view
    - You can see the number of lines or rows by:
      - `awk 'END {print NR}' ${data_loc}/test.ped` `awk` is a very efficient tool for checking text-files
  - .ped: individual genotypes; .map: description of SNPs.
    - They are headerless (no column names).
- PLINK2 genotypes:
  - .psam: sample information (Family ID; Individual ID; Sex); equivalent of .fam in PLINK1.9
  - .pvar: variant information (SNP positions, alleles); equivalent of .bim in PLINK1.9
  - .pgen: genotype data (compressed, efficient); equivalent of binary format .bed in PLINK1.9
  - epihealth.psam, epihealth.pvar, epihealth.pgen

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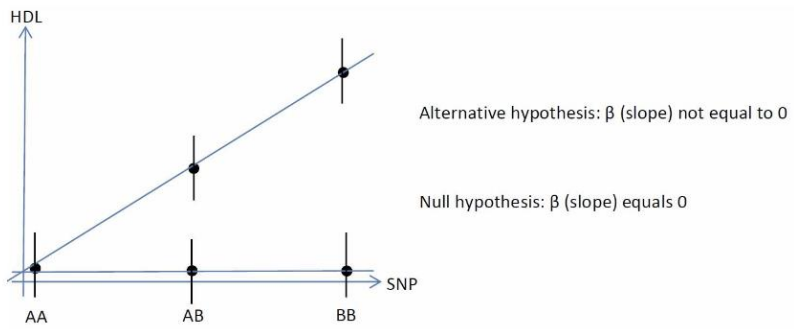
# Instructions for the lab:

https://github.com/YADengUU/Data_Lab_2025

# Association test

- Identify the SNPs associated with HDL levels
- Input data:
  - QC-ed Genotype data created in Lab 1
    - Flags to read files: --pfile (PLINK2); --bfile (PLINK1 binary)
    - If the 3 files don't share the same name, specify by pgen/psam/pvar; bed/bim/fam
  - Phenotype file: epihealth_hdl.pheno
  - Covariate file: epihealth_pcs.covar




Preview the phenotype and covariates to see the names of phenotype and covariates:

```
head ${data_loc}/epihealth_hdl.pheno |
column -t
```

Is it a binary or continuous phenotype?

```
head ${data_loc}/epihealth_pcs.covar |
column -t
```

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# Association test

- --glm hide-covar cols=+a1freq,+ax,+beta
  - --glm is the flag to perform association analysis
  - In the options behind you choose what to show in your results
- --pheno <phenotype-path> Path = directory/filename
- --pheno-name <name-of-phenotype>
- --covar <covariate-path>
- --covar-name <list the covariates, separated by a space in between>
- --out <your customized output name>
- Altogether
  - plink2 --glm hide-covar … --pfile … --pheno … --pheno-name … --covar … --covar-name … --out …

Trick: you don't need the separately created genotype data, the QC flags can be included together in the PLINK2 command line

# Read the results

- As mentioned, awk is a highly efficient tool to check data saved as space/tab-delimited form
  - `awk '$N<5e-8' gwas-result-filename`
  - Replace N with the actual column number containing the p-values
  - If you want to save the quick result: add " `> quick-result-name`" at the end
  - To sort in ascending order, such as for the 3rd-column:
    - `awk '$N<5e-8' gwas-result-filename | sort -gkN > quick-result-name`
    - -gkN: numerically (-g) by the N-th column (-kN)
- Read and visualize data In R:
  - `gwas_result <- read.table("gwas-result", header=TRUE)`
  - `dim(gwas_result)` # check how many rows and columns, if not correct, need to modify sep="\t" or sep=" " in read.table().
  - `library(qqman)` Package "qqman" for visualizing GWAS results

```
png(file="Manhattan-plot.png", width=80, height=50, units="mm", res=300)
manhattan(gwas_result, chr="column-of-chrom-num",bp="column-of-SNP-position", p="column-of-p-val", snp="column-of-SNP-ID")
dev.off()
```

  - Q-Q plot: `qq(gwas_result[, "column-of-p-val"])`

Adjust the figure size in png() before plotting; alternatively, if you have saved the GWAS results in your local computer, you can directly plot without png() or dev.off() and see your figures in RStudio

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# Meta-analysis of GWAS

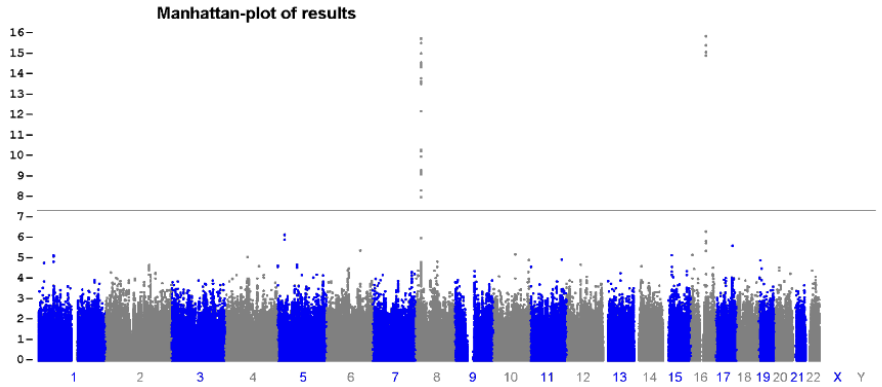
- Most GWAS identify SNPs with small effect sizes and a larger number of statistical tests are performed
- The statistical power to detect a true association between outcome and SNP is low if the sample size is small
- To increase the statistical power, GWAS results from multiple studies are typically combined using meta-analysis

# Meta-analysis in METAL

- In this data lab, the regression results from two studies will be combined using a fixed-effects, inverse-variance weighted meta-analysis
- See instructions for the lab (https://github.com/YADengUU/Data_Lab_2025) for:
  - Create a METAL script file, which is a sample.txt file, using text-editor, not Word.
  - Run the METAL program

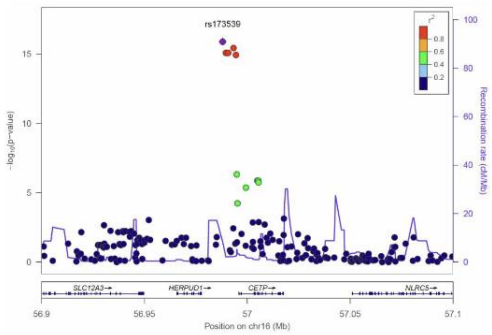
## Plot meta-analysis results genome-wide

- R package qqman
- Input: meta-analysis results file with a chr and position column added




## Or for a selected region

- Use web-based tool Locuszoom
- Meta-analysis results for a selected region around the CETP gene will be used

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